MIL-S-40608D(GL)
28 December 1983
SUPERSEDING
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16 December 1968

MILITARY SPECIFICATION

STOVE, GASOLINE, 2-BURNER

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of portable, 2-burner, gasoline stove with a windshield and carrying case.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-650	-	Brazing Alloy, Copper, Copper-Zinc, and Copper-Phosphorous
QQ-B-654	-	Brazing Alloys, Silver
QQ-C-390	-	Copper Alloy Castings (Including Cast Bar)
QQ-N-281	-	Nickel-Copper-Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7310

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QQ-N-290 - Nickel Plating (Electrodeposited)
QQ-S-571 - Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy
TT-E-485 - Enamel, Semigloss, Rust-Inhibiting
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-843 - Cushioning Material, Cellulosic
PPP-P-291 - Paperboard, Wrapping and Cushioning
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

MILITARY

MIL-L-10547 - Liner, Case, and Sheet Overwrap, Water Vaporproof or Waterproof, Flexible
MIL-W-10727 - Tin Plating; Electrodeposited and Hot-Dipped, for Ferrous and Nonferrous Metals

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

2-9-197 - Stove, Gasoline, 2-Burner, Assembly Complete
2-9-198 - Frame Assembly and Leg
2-9-207 - Spare Parts Holder Assembly and Instruction Plates
2-9-209 - Cover Assembly and Details

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document should take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), before production is commenced, a preproduction sample shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.3. The approval of the preproduction sample authorizes commencement of production, but does not relieve the contractor of responsibility for compliance with all applicable provisions of this document. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Materials (see 6.4) and components. Materials and components not definitely specified shall be of the quality normally used for the purpose in commercial practice.

3.2.1 Castings. Brass castings shall conform to alloy No. A3, type I, II, or III of QQ-C-390.

3.2.2 Blued steel. Blued steel sheet shall be hot rolled or cold rolled, annealed. The sheet shall be oxidized to produce a uniform dark blue color. The steel shall have the following chemical analysis:

Elements	Percent (maximum)
Carbon	0.15
Manganese	0.60
Phosphorous	0.040
Sulphur	0.050

3.2.3 Nickel-copper-alloy. Nickel-copper-alloy wire shall conform to QQ-N-281.

3.2.4 Solder.

3.2.4.1 Brazing. Brazing alloy shall conform to QQ-B-650. Silver solder conforming to QQ-B-654 may be substituted for brazing alloy.

3.2.4.2 Soft solder. Soft solder shall conform to composition Sn 50 of QQ-S-571.

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3.2.5 Plastic. The plastic for the valve knob shall meet all of the applicable requirements of the applicable drawing. The finished knob shall show no sign of deterioration when tested as specified in 4.4.1.1.

3.2.6 Wire cloth. Wire cloth shall be made of either brass wire, or nickel-copper-alloy wire cloth.

3.2.7 Hardware. Hardware items shall be as specified herein and of the grade commonly used in commercial practice.

3.3 Construction. The construction of all components and assemblies of parts covered by this document shall conform to the requirements herein and to the design, details, dimensions, and material classification shown on supporting drawings.

3.3.1 Soldering and brazing. All surfaces to be soldered or brazed shall be thoroughly cleaned prior to soldering or brazing. Soldered and brazed joints shall be smooth, sound, clean, and free from pits.

3.3.2 Burner. The burner head shall be assembled as shown on Drawing 2-9-202.

3.3.3 Wrench. The wrench shall be attached to the stove as shown on Drawing 2-9-197.

3.3.4 Sealing of valve bushing. Thread sealing compound shall be applied to the threads on the valve body prior to assembly of valve in tank.

3.3.5 Lubrication. The air pump leather cup shall be lubricated with thin, silicone-base oil or neatsfoot oil prior to assembly.

3.3.6 Gasoline tank. After assembly, the gasoline tank shall not leak when subjected to the test specified in 4.4.2.2.

3.3.7 Stoves. The completely assembled stove shall have no leaks when tested as specified in 4.4.5 and leaded fuel shall burn with a flame in a stabilized condition, free of carbon when subjected to the operational test specified in 4.4.6.

3.4 Finish.

3.4.1 Electrodeposited nickel plating. All parts specified on drawings to be nickel plated shall be electrodeposited nickel plated 0.004 inch thick in accordance with class 2 of QQ-N-290.

3.4.2 Electrodeposited tin plating. All parts specified on drawings to be tin plated shall be electrodeposited tin plated with a minimum thickness of 0.002 inches conforming to type I of MIL-T-10727.

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3.4.3 Zinc plating. All parts specified on drawings to be zinc plated shall be zinc plated in accordance with commercial practice.

3.4.4 Baked enamel. All parts shown on Drawing 2-9-198 shall be coated with a heat resistant black enamel capable of withstanding 600°F. The gasoline tank shall be coated with a baked enamel conforming to TT-E-485, and conforming to Olive Drab color No. 24087 of FED-STD-595. All parts shall be thoroughly cleaned and degreased before coating with enamel. The thickness of the black enamel coating shall be a minimum of 0.0005 inch. The olive drab enamel coating shall have a minimum thickness of 1.0 mil and shall be baked 45 minutes at 250°F or at a time temperature cycle recommended by the contractor and approved by the contracting officer.

3.5 Spare parts. The following expandable spare parts shall be furnished with each stove:

(a) 2 spare parts holders as shown on Drawing 2-9-207. Each spare parts holder shall contain 1 generator assembly and 1 valve stem packing.

(b) In addition the following shall be furnished in a separate pack:

1 valve stem packing
1 leather compression cup
2 generator assemblies with needles

3.6 Marking.

3.6.1 Instruction plate. Instruction plates or decalcomania shall be as shown on Drawing 2-9-207. The letters for the print shall be characters not less than 1/16 inch high, except the words "TO LIGHT" shall be in upper case characters not less than 5/64 inch high.

3.6.2 Marking for indentification. The letters "U.S." shall be stamped on the frame support and cover in the location shown on the applicable drawings. In addition, the manufacturer's name or trademark, and year of contract shall be shown on the frame support in the location shown on the applicable drawing.

3.7 Workmanship.

3.7.1 Fabrication. All shapr burrs, slivers, or splinters shall be removed.

3.7.2 Welding and brazing. All surfaces of parts to be welded or brazed shall be clean. Welding and brazing shall be accomplished in a manner to prevent the occurrence of burn holes, cracks, fractures, or incomplete fusion. All scale or flux deposits shall be removed from finished welds.

3.7.3 Threaded fasteners. All bolts, nuts, screws, and studs shall be drawn up tight.

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3.8 Finish. The enamel, plating, and blued coating shall be free of areas of no film, blisters, chips, orange peel, scratches, dirt, grit, tacky enamel, foreign matter imbedded in finish, flaking, stained spots, discoloration, tear drops, or grease.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table I, dimensions specified, and tested in accordance with 4.4.5 and 4.4.6. The presence of any defect shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Valve knob testing. The plastic valve knob shall be tested for compliance with the requirement in 3.2.5 by subjecting the knob to a continuous heat of 400°F for 4 hours. Any sign of deterioration shall be considered a test failure. The sample shall consist of three knobs selected at random from the lot. Any test failure shall be cause for rejection of the lot.

4.4.2 In-process inspection.

4.4.2.1 In-process examination. Examination shall be made to establish that all surfaces are cleaned prior to soldering or brazing. Whenever a deviation is noted, correction shall be made to the affected item and lot in process.

3.7.3 Threaded fasteners. All bolts, nuts, screws, and studs shall be drawn up tight.

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4.4.2.2 Gasoline tank assembly leakage testing. Every gasoline tank assembly shall be sealed, filled with air at a pressure of not less than 30 psi, and submerged in clear liquid. Any evidence of failure to conform to the requirement in 3.3.6 shall be cause for rejection of the gasoline tank assembly.

* 4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of stoves. The sample unit shall be one stove. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item defects

Examine	Defect	Classification Major	Minor
Finish:			
Enamel	Coating omitted, not type specified		X
	Blistered, chipped, orange peel, scratches or area of no film		X
	Not finished where required	X	
	Dirt, grit, or foreign matter imbedded in paint		X
	Paint tacky (not dry) or not adherent	X	
Plating	Unplated area	X	
	Not adherent, i.e., peeling, flaking, or blisters	X	
	Stained spots, discolorations, dull areas, tear drops, dirt, grease, or other foreign matter imbedded in finish	X	
Construction and workmanship (general) (applicable to all components and assemblies)	Any component fractured, split, punctured, dented, deteriorated, bowed, or malformed	X	
	Part missing or not specified type	X	
	Any component misplaced, loose, or not in proper alignment	X	
	Sharp burr, sliver, or splinter	X	
	Functioning component that requires abnormal force to operate		X
	Not connected or joined as specified	X	
	Any adjustable assembly that cannot be adjusted to perform the function intended	X	

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TABLE I. End item defects (cont'd)

Examine	Defect	Classification Major	Minor
Soldering and brazing	Missing, not adherent, or cracked	X	
	Incomplete, not clean (flux or flux residue not removed not smooth, surface not neatly finished, loose fins, or particles of brazing compound not removed		X
Bolts, nuts, screws, studs, and other types of threaded fasteners	Broken, stripped, or fractured	X	
	Not type specified		X
Identification marking	Missing, incomplete, illegible		X
Instruction plate	Missing, not specified type or size, not in proper location, not affixed in prescribed manner	X	
Spare parts	Any part missing or not as specified		X

4.4.4 End item dimensional examination. The end item shall be examined for conformance to dimensions specified on the drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of stoves. The sample unit shall be one stove. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.4.5 End item leakage testing. Every stove shall be tested by either of the following methods:

a. With valve closed and orifice needle down, pump 30 strokes. The stove shall then be submerged in a clear liquid and checked for leaks.

b. With valve closed, orifice needle down and filled with air at not less than 30 pounds per square inch (p.s.i.) pressure, all stove connections not previously tested shall be checked for leaks by the application of a liquid leak detector.

Any failure to conform to the leakage requirement in 3.3.7 shall be cause for rejection of the stove.

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4.4.6 End item operational testing. The stove shall be started in strict compliance with the specified operating instructions, operated for a period of not less than 15 minutes, and examined for conformance to the operating requirements in 3.3.7. After the test is completed, all unused gasoline shall be removed from the stove. Any failure to conform to the operating requirements in 3.3.7 shall be classified as a defect. The lot size shall be expressed in stoves. The sample unit shall be one stove. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units shall be 2.5.

* 4.4.7 Packaging inspection. An examination shall be made to determine that preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete sealing or securing of wrap of additional spare parts Bulging or distorted of container
Content	Number of stoves per container is more or less than specified

* 4.4.8 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordane with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps, specified

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Examine	Defect
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified below.

5.1.1 Level A.

5.1.1.1 Spare parts. Spare parts shall be placed in the spare parts holders as shown on Drawing 2-9-207 and the two spare holders placed in the retainers provided on the underside of the stove top. Additional spart parts specified in 3.5 shall be completely wrapped in cushioning material conforming to type II, class B of PPP-C-843, or type I, style 1 or 2 of PPP-P-291 and the wrap secured with tape conforming to type III, grade A of PPP-T-45. Care shall be exercised to insure that tape completely seals the wrap. The wrapped additional spare parts shall then be secured to the stove with tape specified. Tape shall completely encircle the spare parts a minimum of two times.

5.1.1.2 Combination wrench. The combination wrench shall be placed in the retainer provided.

5.1.1.3 Stove. The legs of the stove shall be rotated back until they rest in the grooves in the side of the frame. The pump handle shall be pushed to its limit into the fuel tank pump and the burner valves closed. The stove and the two folded windbreak assemblies shall be placed in the carrying case; the top of the stove positioned facing the carrying case handle, one windbreak assembly placed below and the other above the stove. The upper windbreak panel shall be positioned with its large panel side up. The manual shall be placed on top of the upper windbreak panel. The carrying case cover shall be secured in place with the latches provided.

5.1.1.4 Unit packing. Each stove preserved as specified above, shall be unit packed in a snug-fitting fiberboard box conforming to style FOL, type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. The inside dimensions of the box shall approximate 9-1/2 inches in length, 4-1/2 inches in width, and 16-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Closure shall be in accordance with method II of the appendix of PPP-B-636.

* 5.1.2 Commercial. Each stove shall be preserved in accordane with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified below.

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* 5.2.1 Level A packing. Five stoves, preserved as specified in 5.1, shall be packed in a cleated-plywood shipping container conforming to overseas type of PPP-B-601. Level A unit packs shall be arranged five in length with approximate inside dimensions of the shipping container 24-3/8 inches in length, 17-5/8 inches in width, and 10 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547. Closure and strapping shall be in accordance with PPP-B-601.

* 5.2.2 Level B packing. Five stoves, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or type SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be arranged five in length with approximate inside dimensions of the shipping container 24-3/8 inches in length, 17-5/8 inches in width, and 10 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III as specified in the appendix of the container specification.

* 5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2) the shipping containers shall be V3c, V3s, or V4s, fiberboard box fabricated in accordance with PPP-P-636 and closed in accordance with method III as specified in the appendix of the container specification.

* 5.2.3 Commercial packing. Stoves, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2) stoves packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each coarse. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 and all documents referenced herein or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The stove is intended for use in the field, by groups of from 2 to 5 men, for heating rations and water and melting snow for drinking purposes. The stove may also be used in the field laboratories with laboratory equipment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is not required (see 3.1).

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c. Selection of the applicable levels of preservation and packing (see section 5).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
e. When palletization is required (see 5.3).

* 6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one complete stove. The contracting officer should include specific instructions in all acquisition documents, regarding arrangements for inspection and approval of the first article.

* 6.4 Recylced material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 Changes from previous issue. The margins of this document have been marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

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Review activity:

Army - MD

Preparing activity:

Army - GL

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